squeezed into one another. In that one sentence you will see an enormous amount of information about the world, if just a little imagination and thinking are applied.[10]

Without needing anything from modern physics, Democritus had already arrived at the idea that everything is made up of indivisible particles. How did he do it?

He had arguments based upon observation; for example, he imagined, correctly, that the wearing down of a wheel, or the drying of clothes on a line, could be due to the slow flight of particles of wood or of water. But he also had arguments of a philosophical kind. Let's focus on these, because their potency reaches all the way to quantum gravity.

Democritus observed that matter could not be a continuous whole, because there is something contradictory in the proposition that it should be so. We know of Democritus's reasoning because Aristotle reports it.[11] Imagine, says Democritus, that matter is infinitely divisible, that is to say, it may be broken down an infinite number of times. Imagine then that you break up a piece of matter ad infinitum. What would be left?

Could small particles of extended dimension remain? No, because if this were the case the piece of matter would not yet be broken up to infinity. Therefore, only points without extension would remain. But now let us try to put together the piece of matter starting from these points: by putting together two points without extension you cannot obtain a thing with extension, nor can you with three, or even with four. No matter how many you put together, in fact, you never have extension, because points have no extension. Therefore, we cannot think that matter is made of points without extension, because no matter how many of these we manage to put together, we never obtain something with an extended dimension. The only possibility, Democritus concludes, is that any piece of matter is made up of a *finite* number of discrete pieces which are indivisible, each one having *finite* size: the atoms.

The origin of this subtle mode of argumentation pre- dates Democritus. It comes from the Cilento region in the south of Italy, from a town now called Velia, which in the fifth century BCE was a flourishing Greek colony called Elea. Parmenides lived there, the philosopher who had taken to the letter – perhaps too much – the rationalism of Miletus and the idea born there that reason can reveal to us how things can be other than they appear. Parmenides had explored an avenue to truth via pure reason alone, a path which led him to declare that all appearances are illusory, thus opening a direction of thinking that would progressively move towards metaphysics and distance itself from what would come to be known as 'natural science'. His pupil Zeno, also from Elea, had brought subtle arguments to bear in support of this fundamentalist rationalism, which refutes the credibility of appearances radically. Among